Thiago Pari

347-409-0333 | parimaquera.t@northeastern.edu | linkedin.com/in/thiago-pari | github.com/thiagopari | thiagopari.github.io/ThiagoPari

Education

Northeastern University **Boston, MA**
M.S., Robotics GPA: 3.7/4.0 *May 2027*
Coursework: Computer Vision, Robotic Science and Systems, Mobile Robotics, Robotics Sensing and Navigation, Machine Learning

New York University **New York, NY**
B.A., Economics and Mathematics GPA: 3.6/4.0 *May 2025*
Coursework: Numerical Analysis, Linear Algebra, Differential Geometry, Fourier Analysis, PDEs, Mathematical Modeling

Experience

Archenaux Robotics **Somerville, MA**
Robotics Engineer Co-op, Underwater Perception & 3D Mapping *May 2026 – Sep 2026*
- Owned 3D mapping architecture for a hull ROV; ruled out visual SLAM (median 51 ORB keypoints/frame), moving to sonar + IMU
- Traced sonar SLAM failure (0–6% of motion recovered) to mount bias, not noise; recommended horizontal (0.2–0.3 m ATE in sim)
- Audited 8 pool rosbags and found 4 data defects, including an IMU frame bug (FRD vs. FLU) that reversed lap headings
- Wrote and deployed ROS 2 drivers for the Ping360 sonar and VN-100 IMU; found a serial bottleneck (per-beam time fell $\sim 4\times$)
- Built a hull-inspection segmentation pipeline (partial-label loss); real-hull mIoU 0.12 $\rightarrow$ 0.61, 4.5 FPS on a Jetson Orin Nano
- Built Isaac Sim tools: a domain-randomized Replicator synthetic-data pipeline (biofouling) and a rosbag-calibrated sonar simulator

Citi **New York, NY**
Summer Financial Analyst *Jun 2024 – Aug 2024*
- Automated daily reporting with Excel VBA macros, cutting processing time by 50%

Projects

Underwater SfM Degradation Study | *Python, COLMAP* | *github.com/thiagopari/underwater-sfm-degradation* *Jul – Sep 2026*
- Built a physics-based underwater renderer (measured Jerlov optics, sensor noise) and scored COLMAP poses over 64 seeded runs
- Fitted DPT-SwinV2 monocular depth to COLMAP sparse depth (RANSAC, 82% inliers) to render range-dependent water per pixel
- Showed SfM keeps one map to optical depth 2.4, splits at 2.7, and loses cameras from 3.2; color correction does not move it
- Found SfM fails by dropping cameras, not by bad poses: in 62 of 63 runs every kept camera was within 10 cm and 2°
- Benchmarked ALIKED + LightGlue vs. SIFT in turbid water: $5\times$ the points and 14 vs. 10–12 registered cameras, at $6\times$ the CPU time

Multi-Sensor Fusion for Mobile Robot Localization | *ROS 2, Python, EKF / UKF* *Nov 2025 – Jan 2026*
- Built a 50 Hz ROS 2 EKF fusing wheel odometry, IMU heading, and 1 Hz simulated GPS for a TurtleBot3 in Gazebo
- Benchmarked it against a UKF and dead reckoning: 1.58 m RMS error vs. 2.22 m, a 28.8% reduction
- Simulated 15 s GPS outages; the filter kept tracking on odometry + IMU, with live 3σ covariance ellipses
- Wrote a multi-resolution, trimmed ICP scan matcher for 2D LiDAR: 96% match success, 0.023 m RMSE in standalone tests

12-DOF Quadruped Robot | *Python, Isaac Lab, MuJoCo, Raspberry Pi* *Oct 2025 – Present*
- Built a 12-DOF quadruped on Will Cogley's RoboFox design (Raspberry Pi, IMU, I2C servos) that stood stably
- Modeled its closed-loop legs exactly in MuJoCo (8 loop pins, gear equalities) and as an equivalent tree for Isaac Sim (URDF/USD)
- Trained IMU-only PPO walking policies in Isaac Lab with servo lag, backlash, and gain randomization; no falls in evaluation
- Redesigned the legs with 2:1 gearing in Fusion 360, cutting worst-case servo load from 72% to 40% of stall in simulation
- Wrote the Pi runtime (50 Hz ONNX policy, BNO055 IMU, PCA9685 servos), matched to PyTorch within 4e-7 in sim-in-the-loop tests

Skills

Computer Vision & SLAM: OpenCV, COLMAP / SfM, GTSAM, KISS-ICP / GICP, Open3D point clouds, EKF / UKF sensor fusion
Deep Learning: PyTorch, semantic segmentation, object detection (YOLO), foundation models (DINOv2, AnyLoc), synthetic data
Robotics & Simulation: ROS 2 (rosbag2, DDS, TF, MAVROS), Isaac Sim / Replicator, Gazebo, MuJoCo; sonar, IMU, GPS, cameras
Programming & Tools: Python, C++, MATLAB, SQL, Git, Linux, Docker, pytest, GitHub Actions CI, Claude Code; Fusion 360
Hackathons: Dell $\times$ NVIDIA GB10 (Bayesian skill gate for GR00T VLA robots, team of 4, Oct 2026)
Languages & Interests: English, Spanish; Ironman training, chess (1600 rapid), guitar